A COST BENEFIT ANALYSIS REPORT OF:

HECATE OFFSHORE WIND PROJECT IN BRITISH COLUMBIA

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EXECUTIVE SUMMARY

This Cost-Benefit analysis is intended to scrutinize the Hecate offshore wind farm located in British Columbia, with a capacity of 400 MW. Our study will be an In-Media-Res analysis since the project is ongoing though still in the initial processing stage. We will evaluate the economic viability and potential benefits and costs of the project to the province, considering both the costs involved in its implementation and the expected returns over its operational lifespan.

The study elucidates the primary purpose of the project and draws comparisons with other types of power plants.The analysis aims to determine the impacts on provincial standings, from the Cost-Benefit Analyst's perspective. We will quantify and monetize the various costs associated with the offshore wind farm's construction, installation, and maintenance. These costs may include expected capital expenditures, operational expenses, transmission infrastructure, and environmental impact mitigation measures. These costs will be estimated, as the cost of construction depends on the prices offered by contractors during the bidding process.

Next, the analysis will identify the potential benefits of the Hecate offshore wind farm. This will involve estimating the wind farm's electricity generation capacity and calculating the expected revenue from selling the generated electricity. Additionally, the analysis will consider potential environmental benefits, such as the reduction of greenhouse gas emissions. It will also examine any social or community benefits that may arise from the project. To assess the economic viability of the Hecate offshore wind farm, the analysis will employ various financial metrics, including net present value (NPV) and a real social discount rate of 3.5% covering the project's 25-year lifespan (since the project will have an impact less than 50 years)[1].

[1] These values are based on those laid out in Mark A. Moore, Anthony E. Boardman, Aidan R. Vining, David L. Weimer, and David H. Greenberg, "Just Give Me a Number! Practical Values for the Social Discount Rate." Journal of Policy Analysis and Management, 23(4), 2004, 789–812.

Additionally, the analysis will consider potential risks and uncertainties associated with the project, such as fluctuations in electricity prices, regulatory changes, and technological advancements. A sensitivity analysis will be conducted to evaluate the project's resilience to these uncertainties and identify critical factors that may significantly impact its financial viability.

The findings of this cost-benefit analysis will provide valuable insights for stakeholders, policymakers, and investors interested in the Hecate offshore wind farm. The results will keep them informed in good decision-making processes and facilitate a comprehensive understanding of the project's economic feasibility and potential benefits.

CHAPTER 1

INTRODUCTION

1.1 Background to the Study

Cost-Benefit Analysis provides a structured way to gauge policies, programs, and projects by measuring their impact in monetary terms for society as a whole. This systematic method is widely used in business investments and public policies to weigh the potential costs and benefits of proposed decisions or projects. The main goal of Cost-Benefit Analysis (CBA) is to determine if the benefits of a decision outweigh its costs and if it ultimately generates a higher net social benefit. In this project report, we will conduct a cost-benefit analysis of the Hecate Offshore Wind Farm project located in British Columbia. With a capacity of 400 MW, this project holds the potential to become one of the province's leading sustainable energy sources.

1.2 Statement of the Research Problem

- Why choose an offshore wind project over other energy generation sources?
- Is the prevalent use of hydropower sources in BC environmentally sustainable?
- What are the benefits and costs of implementing the offshore wind project?
- Do alternative projects present similar advantages to this proposed project?
- Will the residents of British Columbia find value in adopting this energy source?

Finding theoretical and empirical answers to these questions is what motivates and makes our study an interesting one.

This research aims to explore how an offshore wind project, such as this one, could alter the landscape of electricity generation in British Columbia and its societal implications.

1.3 Objectives of the Study

The primary objective is to assess the costs and benefits of the Hecate Offshore Wind Farm Project, analysing its societal impact in monetary terms and propose recommendations for decision-makers based on available information.

The specific aims include:

1. Evaluating the short and long-term advantages of the Hecate Offshore Wind Farm Project.
2. Analyzing the project's environmental effects on society.
3. Comparing the Hecate Offshore Project against alternative options, considering their operational lifespans.
4. Offering valuable recommendations from our analysis and findings to stakeholders, policymakers, and investors interested in the Hecate OffshoreWind Farm.

1.4 Significance of the Study

The significance of the study lies in the uniqueness of the Hecate Offshore Wind Project; upon its completion, it will become the first of its kind in British Columbia. Hence, using the cost-benefit analysis as an economic tool boosts the significance of this research, aiming to reveal the factors impacting the province's energy sources, both positive and negative, for society. Our research will provide stakeholders, policymakers, and investors with information for those interested in offshore wind farms like this project.

1.5 Scope of Study

Due to the difficulty in accessing relevant data, the study is confined to analyzing the benefits and costs of an offshore wind farm, specifically centered on the Hecate Offshore Wind Farm in British Columbia, constructed between the Queen Charlotte Islands and Vancouver Island. The nearest metropolis to the project area is Vancouver, with an estimated population of about 2.71 million in 2023.

The study will also look at some economic and financial metrics in our evaluation to help us in assessing the project's impacts on the people of British Columbia. Data on impacts were collated from various secondary sources related to our project, as described in the third chapter of the study.

1.6 Justification of the Study

The rationale behind provinces like British Columbia undertaking offshore wind projects lies in its decision to foster rapid economic development through environmentally conscious methods. Drawing from the experiences of other nations struggling with recurring economic crises, particularly those caused by climate change, there's a pressing need for enhanced measures to uphold a positive balance in the energy sector. Which will therefore serve as a pathway to achieve sustainable economic growth. However, these beliefs need empirical investigation to validate policy recommendations. Thus, the study endeavors to explore how this equilibrium will tangibly benefit British Columbia's society.

1.7 Organization of the Study

The study will be organized into five main chapters. The chapters are as detailed below:

Chapter One: This chapter will introduce the study under the following headings: background of the study, statement of the research problem, objective of the study, significance of the study, the scope of the study, justification of the study, and the organization of the study.

Chapter Two: This chapter concentrates on the review of the relevant theoretical and empirical literature on our cost-benefit analysis of the Hecate Offshore Wind Project and information on other energy-related projects.

Chapter Three: The chapter will discuss the methodology of the study. This will include the following: focus on the research method or approach being used in the study, a brief description of the step-by-step approach utilized in the study, and the research design.

Chapter Four: This chapter will analyze, interpret, and present the collected data.

Chapter Five: This will be the final chapter and will therefore give the summary of the study, the conclusion of the study, and recommendations respectively.

CHAPTER 2

LITERATURE REVIEW

2.0 Introduction

This chapter presents a review of the theoretical and empirical literature on our cost-benefit analysis of the Hecate Offshore Wind Farm, taking into consideration some relevant information derived from other wind energy-related projects. The chapter is organized in three main sections. The first section is devoted to the definitions of some key terms that are frequently used in this study. The second part encompasses the theoretical literature, and the third part shows a summary of the empirical literature on the impact of the project on society.

2.1 Definition of Concepts

2.1.1 Definition of Offshore Wind Farm

According to the Oxford Learners Dictionary, the word 'Offshore' means happening or existing in the sea, not far from the land, and a ‘Wind-farm' is an area of land on which there are a lot of windmills or wind turbines for producing electricity. Hence, it can be concluded that an offshore wind farm is an area situated in the sea, not far from the land, where a multitude of windmills or wind turbines are installed to generate electricity.

2.1.2 Definition of Onshore Wind Farm

An onshore wind farm is an area populated with numerous windmills or wind turbines designed to generate electricity (Oxford’s Learners Dictionary).

2.1.3 Definition of Hydro Power Plant

Hydroelectric power is electricity produced by generators driven by turbines that convert the potential energy of falling or fast-flowing water into mechanical energy (Britannica, 2023)

2.1.4 Definition of Thermal Power Plant

This is a power-generating plant that uses heat to produce energy. Such plants may burn fossil fuels or use nuclear energy to produce the necessary thermal energy. (European Environment Agency, 2000)

2.1.5 Definition of Wind Turbines

According to the Office of Energy Efficiency & Renewable Energy US, a wind turbine transforms wind power into electricity by harnessing the aerodynamic force produced by its rotor blades, similar to an airplane wing or helicopter rotor. As wind flows across these blades, it causes a difference in air pressure, resulting in both lift and drag forces. The lift force outweighs the drag, initiating the rotation of the rotor. Connected to a generator, this rotor rotation, either directly or via a gearbox that enhances speed, produces electricity from the aerodynamic power.

2.1.6 Definition of Decommissioning Cost

The definition of decommissioning cost by Oxford University Press are costs incurred in ceasing an operation or activity. An example might be the removal of an oil rig and the restoration of the seabed. Under Financial Reporting Standard 12, a provision should be made for decommissioning costs relating to the damage already done, e.g. the damage caused by the installation of a rig. The relevant International Accounting Standard is IAS 37.

2.1.7 Definition of Net Present Value

Net present value (NPV) is used to calculate the current value of a future stream of payments from a company, project, or investment. To calculate the NPV, you need to estimate the timing and amount of future cash flows and pick a discount rate equal to the minimum acceptable rate of return. (Investopedia).

2.1 8 Definition of Discounting Rate

The discount rate, as defined by Investopedia, is the interest rate used to calculate the present value of future cash flows from a project or investment. Many companies calculate their WACC and use it as their discount rate when budgeting for a new project.

2.2 Theoretical Review

The literature surrounding the analysis of the Hecate Offshore Wind Project in British Columbia encompasses a diverse range of sources, each contributing valuable insights into various aspects of wind power, project economics, environmental impacts, and regulatory considerations. Evidently, in conducting our analysis we relied on the textbook “Cost-Benefit Analysis” 5th edition by Anthony E. Boardman, David H. Greenberg, Aidan R. Vining, David L. Weimer to follow all the steps of CBA and concepts to determine the impacts. Additionally, the following review synthesizes key findings from the selected references, shedding light on critical factors influencing the cost-benefit analysis of the offshore wind project;

1. This article provides a comprehensive comparison between onshore and offshore wind power. It emphasizes that while onshore wind is currently more cost-effective, offshore wind's future prospects are promising due to decreasing costs and land scarcity issues (Beauchamp, 2023).

2. Examining the costs and worth of wind turbines in 2023, this source explores aspects like expenses, electricity production, maintenance, and industry concerns about decreasing prices. It offers insights into turbine sizes, operational costs, and technological advancements (Blewett, 2021).

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3. How to Calculate Construction Labor Cost - This article guides the efficient calculation of construction labor costs, emphasizing the importance of labor rates, burden calculations, and various fees. It provides essential insights into managing costs in construction projects (Bridgit, n.d.)

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4. Provincial Inventory of Greenhouse Gas Emissions - Summarizing the annual Provincial Inventory of Greenhouse Gas Emissions in British Columbia, this source offers crucial data on emissions from 1990 onward, aiding in understanding the environmental impact of the offshore wind project (Government of British Columbia).

5. Rystad Energy: Less Is More if Using 14 MW Turbines - Comparing the financial aspects of different offshore wind turbines, this article suggests that larger 14 MW turbines, despite higher initial costs, result in considerable savings, especially in installation and cabling expenses (Buljan, 2020).

6. Virginia Tech Siting & Project Development Written ReportpmentReport_2019-04-21.pdf) - Focusing on the Wills Ridge wind farm project, this report discusses its financial viability challenges, emphasizing potential solutions like leveraging tax credits and reducing turbine costs (Burger et al., 2019).

7. Electricity rates - This source outlines residential and commercial electricity rates, providing critical information for evaluating the economic aspects of the offshore wind project (FortisBC, n.d.).

8. Operation and maintenance costs of wind-generated power - Examining operation and maintenance costs, this article details the expenses associated with wind turbine upkeep, covering insurance, repairs, spare parts, and administrative costs (The European Wind Energy Association, 2010).

9. Evaluating Offshore Wind Energy Feasibility off the California Central Coast - Assessing offshore wind feasibility, this project involves stakeholder analysis, regulatory review, site mapping, and a benefit-cost analysis, providing insights into financial implications and future research areas (Feinberg et al., 2014).

10. Assessing the expansion of the project- The article emphasizes wind energy's expansion, technological progress, and reduced costs. It highlights the substantial growth in installed capacity, notably offshore, mentioning the increased size of modern turbines onshore and offshore (United States Energy Information Administration, 2023).

These selected references collectively contribute to a comprehensive theoretical literature on the understanding of various factors influencing the Hecate Offshore Wind Project in British Columbia, offering valuable perspectives on costs, environmental impacts, and regulatory considerations.

2.3 Empirical Review

This section reviews the empirical studies relating to offshore wind farm projects and other energy sources. Due to the significance of this study, several empirical studies have been conducted to assess energy sources like wind farms of which we had to use some of that information in our study.

In gathering values for other hydropower stations like the Revelstoke Generating Station (2,480 MW), Mica Generating Station (2,746 MW), and Seven Mile Generating Station (805 MW). B.C. which is home to roughly 16,000 MW of hydroelectric capacity, aided our analysis together with other significant empirical values (Moore, Boardman, Vining, Weimer, & Greenberg, 2004).

We had to take into consideration the expenses associated with wind turbines, detailing initial costs influenced by materials, labor, and location. It breaks down costs for manufacturing, installation, maintenance, and transportation. Overall, it underscores the long-term benefits of wind power while highlighting its environmental advantages and the necessity of renewable energy investment (Duval, 2023).

One of the cost components that cannot be looked down on is the lease period for occupying the space for a project, whether land or sea. Hence, we had to look at some articles that explore leasehold ownership in Canada, focusing on land lease houses and the types of land available for leasing in British Columbia. It details the advantages and disadvantages of leasehold homes, emphasizing cost considerations, property rights, and land lease tenure lengths (WOWA, 2023).

Modeling the decommissioning cost of offshore wind development on the U.S. Outer Continental Shelf was the basis on which we could look since we included that as part of our costs. So, we reviewed a paper that estimates the decommissioning costs for U.S. offshore wind farms, focusing on factors like removal, processing, and disposal expenses. Estimated costs range from $115,000 to $135,000 per MW, comprising about 3-4% of the total capital costs for these projects (Kaiser et al., 2012).

Consider some significant articles that outline British Columbia's Provincial Sales Tax (PST) at a rate of 7% for most goods and services, although there are certain exemptions (Government of British Columbia, 2023).

A good comparison gave us this piece, which delves into the comparison between onshore and offshore wind farms, discussing their advantages and drawbacks. It explores cost-effectiveness, installation speed, economic impact, and environmental considerations for both types. Ultimately, it weighs the pros and cons, foreseeing offshore wind as a potential future alternative despite its current higher expenses and maintenance challenges (Timblin, 2022).

We could not ignore the piece that explores the social cost of carbon, indicating that each tonne of carbon creates $250 in societal damage. It delves into policy implications and the need to align carbon pricing with this cost, emphasizing the unsustainability of certain lifestyle choices (Hugo, 2023).

Additionally, we had to also consider mortality risk. The paper looks at how Canadian policy makers assess lower mortality risk. It explores willingness-to-pay (WTP) metrics that aid in estimating the cost-benefit of programs, such as the value of statistical life (VSL). It highlights how difficult these estimations are to understand and how much more study is required in this area (Lauraine et al., 2009).

This article also about the Skipjack Wind 2 Offshore Wind Farm, a 760 MW project developed by Orsted US Offshore Wind and scheduled to be completed in Maryland, USA in 2027, with an estimated cost of $3.7 billion, is one of the similar projects we reviewed. The project is expected to power 250,000 households (Power Technology, 2021).

The cost of living in Canada and the USA is compared, and topics including healthcare, taxes, employment opportunities, and living standards are covered. It notes that while housing and consumer expenditures are typically lower in Canada, the USA frequently offers greater salaries (Spring Financial, 2023).

Moreover, the financial benefits of 14 MW offshore wind turbines are compared in the article with those of smaller 10 MW and 12 MW units. It demonstrates how using 14 MW turbines saves a significant amount of money for large-scale projects, even with their more expensive production. Additionally, it emphasizes how offshore wind farms are now more financially feasible overall due to decreased foundation and installation costs (Windfair, 2020).

Furthermore, taking into account this article which examines the Levelized Cost of Energy (LCOE) for both land-based and offshore wind power plants in the United States, using 2021 data. It aims to understand cost variability and includes estimated LCOE for various wind projects, and sensitivity analyses (Stehly et al., 2021).

The article explores the complex decisions involved in decommissioning offshore wind farms nearing the end of their lifespan. It outlines three main options: extending the life, upgrading components (repowering), or complete decommissioning (Thomson Environmental Consultants, 2021).

These articles collectively contribute to a comprehensive empirical literature on the understanding of various data sources, considerations, and approaches we employed in our cost benefit analysis for the Hecate Offshore Wind Project in British Columbia, offering valuable perspectives on costs, environmental impacts, and regulatory considerations.

CHAPTER 3

METHODOLOGY

This CBA on the Hecate Wind Farm project is conducted from a Cost Benefit Analyst perspective and focuses on how it can influence the social welfare of British Columbia. In so doing, our methodology for this project is to use the required ten (10) steps of CBA for an effective and efficient analysis. They consist of the following:

Step 1. Purpose of the Project:

The project's main objectives are detailed, encompassing specific information about the project, such as its location, capacity, and key metrics. A significant aspect involves comparing the offshore wind farm with hydroelectric and gas power plants, giving a broader context. Additionally, the approach considers approximate construction costs, acknowledging the practical challenges due to dependencies in the bidding process.

Step 2. Set of Alternatives:

This step involves examining various alternatives, such as onshore wind farms and the status quo, to offer a comprehensive analysis.

The comparison focuses on crucial elements like construction expenses, output, and land costs, providing essential insights into each project's vital factors.

Step 3. Standings:

This step analyses whose benefits and costs count. It clearly identifies primary beneficiaries, and cost-bearers while acknowledging the environmental impacts. This analysis is limited to the provincial perspective, showing awareness of scope limitations.

Step 4. Impact Categories and Catalogue:

Here, we examined the costs and benefits, covering wind turbines, transportation, installation, electrical infrastructure, land use, miscellaneous expenses, operation and maintenance, decommissioning, revenue from power generation, social cost of carbon, and scrap value.

Step 5. Predict the Impacts Quantitatively Over the Life of the Project:

Assess the effects encompassing wind turbines, transportation, installation, electrical infrastructure, land utilization, decommissioning, ongoing maintenance, power generation revenue, social cost of carbon, and scrap value over the entirety of the project's lifespan.

Step 6. Monetizing Impacts:

Assign monetary values to impacts, considering costs (wind turbines, transportation, installation, electrical infrastructure, land, decommissioning, miscellaneous, operation and maintenance) and benefits (revenue from electricity generation, social cost of carbon, scrap value).

Step 7. Discounting:

In Step 7, the present values of the impacts over the project horizon were calculated by applying a discounting rate of 3.5%. The impacts were categorized into Initial Capital, Annuities, and Terminal Values.

Step 8. Compute the net present value of each alternative:

The NPV (Net Present Value) was calculated by subtracting total costs from total benefits after discounting and it was done for the alternative project as well.

Step 9. Sensitivity Analysis:

The sensitivity analysis considered here is the best and worst-case scenario. Construction costs were scrutinized, and the Levelized Cost of Energy (LCOE) was calculated to assess the accuracy of the construction cost estimation. The Internal Rate of Return for both scenarios was also determined.

Step 10. Recommendations:

After the calculations for the NPV and Sensitivity analysis of the project, we determined the robustness of the project and provided our recommendations. Our decision rule will be that if NPV is positive (NPV > 0), implement the project, but if NPV is negative (NPV < 0) then, do not implement the project. Overall, with respect to alternative projects, the project with the highest NPV will be most preferred.

Overall Evaluation:

The methodology is robust, well-structured, and aligns with best practices in cost-benefit analysis. It incorporates a comprehensive literature review, considers alternative scenarios, and outlines a clear plan for analysis, including sensitivity testing. The recognition of potential uncertainties and the focus on practical decision-making make this methodology thorough and applicable to real-world scenarios.

Step 1. Purpose of the project

The primary aim of this Cost-Benefit Analysis on the Hecate Offshore Wind Farm, currently under development, is to aid in effective decision-making, ascertain the Net Present Value, and overall social benefit while assessing its financial and environmental implications. Additionally, it aims to compare the project's efficiency and viability against alternative options. As an ongoing initiative, this analysis anticipates the potential outcomes of the project. The basic data available for assessing this project includes its location, capacity, number of turbines, and its owner. The wind farm will be constructed in the sea between the Queen Charlotte Islands and Vancouver Island. The nearest metropolis to the project area is Vancouver, with an estimated population of 2.71 million in 2023, according to PopulationU.com. The total capacity of the wind farm is expected to be 400 MW, which can generate 3,504,000 MWh annually. The average electricity consumption in BC per household is 11,000 kWh annually, as per www.cer-rec.gc.ca, meaning the wind farm can provide electricity to approximately 318,545 households. A total of 33 wind turbines are set to be installed to generate this electricity, indicating that each turbine must have a capacity of more than 12 MW. The owner of the wind farm is Northland Power, Inc. Additionally, the area it will cover is 51.0194 km² and it will be situated 9 km from the shore.

It should be noted that 87% of the province's electricity is generated from hydroelectric sources, as reported by www.cer-rec.gc.ca. This is largely due to the abundance of rivers in Canada; hence, this fact makes this CBA unique since most similar projects' by Analysts are conducted in comparison to fossil fuel-based power plants.

In this context, the province's total electricity generation capacity currently stands at 18,250 MW. In our analysis, we will primarily focus on the benefits of wind power in comparison to gas power plants which are also called thermal power plants. It should be noted that most of the thermal power plants are used during peak demand seasons especially in winter time. The wind farm can substitute these carbon emitting power plants to prevent pollution.

Currently, five natural gas power plants are in use in British Columbia with a total capacity of 507.8 MWh. As a result, Hecate OWF can offset 78.77 % of carbon emissions in the province stemming from electricity generation.

Furthermore, we will calculate other associated costs, such as maintenance and connection to the national grid. It must be acknowledged that the calculation of construction and installation costs will be based on approximate numbers derived from similar projects, as the costs of large projects depend on contractors' offers during the bidding process.

The CBA's horizon is set for 25 years, which aligns with the typical longevity of wind turbines. We will apply a 3.5% real social discount rate since the project has an impact less than 50 years[2].

Step 2. Set of alternatives

As mentioned above, British Columbia is rich in hydroelectricity sources thanks to its vast rivers and mega dams, such as the Revelstoke Generating Station (2,480 MW), Mica Generating Station (2,746 MW), and Seven Mile Generating Station (805 MW). B.C. is home to roughly 16,000 MW of hydroelectric capacity, most of which is located on the Columbia River in south-eastern B.C, and the Peace River in the northeast. Apart from hydropower, 5% of BC's electricity is generated by biomass, geothermal sources, and 4% by natural gas. This represents the status quo, the initial alternative, depicting the original state before any action or project, with a net present value of zero.

It's worth noting that the horizon of the hydropower plant is much longer than that of the wind farms and requires bigger investment to construct giant dams.

Ideally, an alternative project will be an on-shore wind farm which has a similar lifespan, technology and energy source. The differences can be observed in construction costs, since on-shore it is easier to carry out, the construction works and substructures will cost less. Moreover, the output levels might also be different, since in the sea the intensity of wind is much higher thereby generating more power. Also, the cost of land will vary significantly.

[2] These values are based on those laid out in Mark A. Moore, Anthony E. Boardman, Aidan R. Vining, David L. Weimer, and David H. Greenberg, "Just Give Me a Number! Practical Values for the Social Discount Rate." Journal of Policy Analysis and Management, 23(4), 2004, 789–812.

Step 3. Standings

The primary beneficiaries and cost-bearers identified in this step are the developer, the BC government, and the residents. The electricity generated by the project primarily serves the residents of British Columbia, making their standing the utmost priority. While the project could potentially reduce carbon dioxide emissions and aid in the decommissioning of gas power plants, its impact at a global level remains indistinct and challenging to evaluate comprehensively. Hence, our analysis will focus solely on the provincial perspective.

Step 4. Impact Categories and Catalogue

To begin with, let's identify the costs and benefits of the project. As mentioned earlier, it's impossible to calculate the exact cost of the construction and installation works because projects are implemented by contractors whose prices vary during the bidding process. However, we can calculate some primary costs, such as wind turbines, undersea electricity transmission line cables, installation costs, and other miscellaneous expenses. We will also include the annual cost of maintenance and operation, as well as decommissioning costs at the end of the project.

Regarding the benefits, we have taken into consideration several factors, including the annual revenue generated from selling electricity, revenue from selling renewable certificates for the developer, and the scrap value of the equipment at the end of the project. For the government, it involves saving on fossil fuels that can be either exported or used for the internal market, addressing deficiencies in certain regions, and leasing land. For residents, there is the social cost of carbon, which is offset by decommissioning fossil fuel-based power plants, as well as the cost of lives saved due to the reduction in carbon emissions.

Let's elucidate each cost and benefit together with their detailed explanation.

COSTS

Cost of wind turbines, its substructure and foundation.

Turbines with their substructure and foundation are primary components of the wind turbine mechanism which generates the electricity. There is a small chance that the price might change in the market due to small addition to demand in the market. Therefore, the price reflects the social cost of the input.

We will measure the cost of wind turbines in accordance with the number of wind turbines that will be installed depending on the electricity generation capacity of the wind turbines.

Transportation costs.

Transportation costs are related to transporting the equipment to the project's site. The transportation costs mainly depend on the location of the manufacturer. Since most wind turbine manufacturers are based in the United States, it is evident that they will be imported from the US. Due to the substantial costs associated with transporting the massive blades and towers of these turbines, it is rational to select the nearest manufacturer with a good track record. In this regard, EDF Renewables, located in California, can be considered as the turbine supplier, as other major manufacturers are situated in the Eastern part of the continent, which could significantly escalate transportation costs.

It must be admitted that the transportation costs does not solely depend on the distance and weight of the spare parts to be brought to the site. Hence, will give a lump sum value to this impact in comparison with similar projects.

Installation Costs.

Installation costs comprise a notable share of the construction costs of wind farm projects due to a huge size of the turbines and substructures. On top of all, all the work will be carried out in the sea requiring special equipment and mechanisms. Apart from the direct costs we will consider the potential harm to aquatic life and risk to a human life to reflect the true social cost.

Installation costs depend on the number of wind turbines to be installed. Hence our measurement unit will be just the number of wind turbines.

Cost of electrical infrastructure (Grid Connection)

According to the feasibility study of CalWind on Offshore Wind Energy Development on the California Central Coast[3] The necessary electrical infrastructure for the project consists of turbine interconnection cables, a substation, and transmission cables. Turbine interconnection cables play a crucial role in transferring power from each generator to the charge controller or battery in wind turbines. These cables enable the turbines to be easily connected or disconnected from the charge controller using a quick switch.

There is a small chance that the project might affect the market prices and there is no data regarding the supply curve of electrical infrastructure materials in BC. Therefore, we will identify the whole expenditure as a social cost.

It must be noted that the distance of the offshore wind farm from the shore is 9 km. However, it is not primarily about the distance which determines the cost of electrical infrastructure. The number of wind turbines are a vital factor to reveal the cost of the impact. Hence, we will measure this impact proportionally to the cost of wind turbines.

Cost of land.

It's important to note that there is no single offshore wind farm in Canada to determine the precise non-use value of offshore land. Furthermore, there is no data available to identify other alternative uses of the offshore land. It’s rational to accept that the opportunity cost of the land is zero. However, we found that in some

[3] https://tethys.pnnl.gov/sites/default/files/publications/Appendix-D-CalWindProject.pdf

other projects outside of Canada the developer had to lease the land, which gives a sense that there might be some costs associated with usage of the offshore land. We can assume that the project can prevent some fishers from using that sea site for fishing, and cargo ships will have to circumvent the project's site due to safety reasons increasing their transportation costs.

It must be noted that the cost depends on the area of land to be covered by the project. Therefore we will use km2 to calculate the measure of impact.

Miscellaneous Costs

For miscellaneous costs, we are including development, insurance during construction, construction financing, contingency, plant commissioning according to the review of cost of wind energy by Tyler Stehly and Patrick Duffy in 2021[4]

We will measure this impact relative to the total project expenses, making it a lump sum value.

Cost of operation and maintenance

Operation and maintenance (O&M) costs constitute a sizable share of the total annual costs of a wind turbine. O&M costs are related to a limited number of cost components, including:

- Insurance;
- Regular maintenance;
- Repair;
- Spare parts, and
- Administration.[5]

Due to intricacies in the calculation of each component of the operation and maintenance costs, we relied on the data that provided a simplified approach for the O&M costs calculation. According to data from "Wind

[4] https://www.nrel.gov/docs/fy23osti/84774.pdf
[5] https://www.wind-energy-the-facts.org/operation-and-maintenance-costs-of-wind-generated-power.html

Measurement International", for modern machines, the estimated maintenance costs range between 1.5% to 2% of the original investment per annum.[6]

Decommissioning Cost

Decommissioning the wind farm is a fundamental aspect of cost considerations. However, questions may arise regarding its necessity and why leaving it as it is after the project's horizon isn't a viable option. So the main reason is all wind farms will reach the point where the cost of maintenance or repowering will not be commercially viable compared with the cost of decommissioning. Decommissioning involves dismantling the turbines and removing the structure. [7]

The measurement of the decommissioning cost is derived from the capacity of the wind farm. Hence, MWh is our measurement unit.

Environmental Externalities

It's worth noting that offshore wind energy is mainly associated with environmental friendliness, such as the insignificant emission of carbon and other hazardous gases. Furthermore, it doesn't pose barriers to the migration of fish and doesn't overflood lands nearby, which is the primary concern with hydropower. However, there may be potential environmental externalities and risks to aquatic life. Wind farms can cause damage to the seabed and sea creatures during the construction and installation process.

The potential impact of the offshore wind farm has not been fully studied yet and requires several years to conduct a comprehensive Environmental Impact Assessment (EIA). Nevertheless, according to the article "Environmental Impacts of Offshore Wind Power Production in the North Sea" by Nina Jensen, CEO of WWF-Norway, these risks can be mitigated by taking the necessary measures.

[6] http://www.windmeasurementinternational.com/wind-turbines/om-turbines.php
[7] https://www.thomsonec.com/news/offshore-wind-decommissioning/#:~:text=Eventually%2C%20all%20windfarms%20will%20reach,turbines%20and%20removing%20the%20structure.

With that said, we will assume that the environmental externalities of the Hecate project are minimal (a score of 0) and will be considered potential liabilities in the sensitivity analysis.

According to Catherine Bowes, from the National Wildlife Federation she praised Equinor, one of the world's largest offshore windfarm projects in Norway for committing to a new construction technology that will lower enormous prefabricated cement foundations for the wind turbines rather than pile-driving into bedrock to hold the 850-foot-tall steel towers in place. Whales are extremely sensitive to noise, she says, so avoiding the extremely noisy process of pile-driving is a big step. In our project, we took into consideration some of these necessary measures in the foundational construction process.

BENEFITS

Revenue from electricity generation

Evidently, the main output of the wind farm is electricity, which will be sold to consumers in the province. The transmission of electricity is focused on Vancouver Island, which is the nearest populated area. The impact on price due to increased supply is minimal, considering BC's electricity supply of 18,250 MW, in contrast to the Hecate project's 400 MW capacity. Additionally, the project aims to replace thermal power plants utilized during peak demand periods.

Moreover, since wind energy is considered a clean energy and can cause shutting down of the thermal power plants which emit greenhouse gases, we account for this positive externality as a social benefit. But, we will calculate this effect separately as a social cost of carbon.

The unit of measurement for this impact remains KWh per year.

Social Cost of Carbon

The social cost of carbon evaluates the reduction in environmental and human health damage due to the decreased use of natural gas for electricity generation. By opting for "clean" wind energy, the dependency on natural gas-driven power diminishes, resulting in societal benefits. It's important to note that despite its categorization as a social cost, we're treating it as an indirect benefit in our analysis.

We will measure the magnitude of this impact in tonnes of CO_2 prevented from emission.

Scrap value of equipment and materials at the end of the project

Considering the valuable materials used to manufacture powerful wind turbines, it is logical to expect that they will have a scrap value at the end of the project, after decommissioning the wind farm.

Scrap value constitutes a fraction of the wind turbine's value, that’s why it will be measured in the number of wind turbines.

Let’s now catalogue these impacts on the table below.

Types of impact	Provincial perspective			
	Costs	**Unit**	**Benefits**	**Unit**
Initial Inputs	Wind turbines, its substructure and foundation	Nr		
	Transportation	Lump sum		
	Installation	Nr		
	Electrical Infrastructure	Nr		
	Miscellaneous cost	Lump sum		
Annuities	Cost of land	km2	Revenue from electricity generation	KW/year
	Operation and maintenance	Lump sum	Social Cost of Carbon	Tonnes / year
Terminal Values	Decommissioning cost (At the end of the project's horizon)	MWh	Scrap value of equipment and materials at the end of the project	Nr

CHAPTER 4

FINDINGS AND ANALYSIS OF DATA

Step 5. Predict the impacts quantitatively over the life of the project

COSTS

Cost of wind turbines, its substructure and foundation.

Large wind turbines built for onshore and offshore wind farms can generate about 2 to 3 MW, while the largest offshore turbines can generate up to 12 MW of electricity[8]. Considering that 33 wind turbines will be installed to generate 400 MW, the project requires using the largest wind turbines.

Transportation costs.

Transportation cost of wind turbines can be categorized into two types: short distance and long distance transportation. It's highly likely that all the wind turbines will be imported from the USA, which makes it long-distance transportation. Transportation costs will be calculated in accordance with the number of wind turbines for a long distance. As specified above, the number of wind turbines are 33.

Installation Costs.

As we outlined in the previous step the installation costs will be calculated proportionally to the number of wind turbines.

Cost of electrical infrastructure (Grid Connection)

Due to a lack of data on the price of each component of electrical infrastructure, we are extracting it from the Levelized Cost of Energy, which breaks down the cost of electricity. According to a review of cost of

[8] https://www.irena.org/Energy-Transition/Technology/Wind-energy

wind energy by Tyler Stehly and Patrick Duffy in 2021, the cost of electrical infrastructure hovers around 53% of wind turbines' cost[9]. We will multiply it to the number of wind turbines.

Cost of land.

As we outlined in step 4, the assessment of land costs will be based on a comparative scale involving a project of similar scope and land coverage. The offshore area designated for this project spans approximately 55 km2.

Decommissioning Cost

The wind farm has a capacity of 400 MWh, and we will determine the decommissioning cost by multiplying this capacity by the previously calculated cost per MWh from a credible data source.

Miscellaneous Costs

Given the complexities involved in computing these expenses, our estimation indicates that they collectively account for 17% of the overall expenses, encompassing wind turbines, substructure, transportation, installation, electric infrastructure, operation and maintenance, and decommissioning costs which we derived from the Review of Cost of Wind Energy by Tyler Stehly and Patrick Duffy in 2021[10]

Cost of operation and maintenance

As highlighted in step 4, the yearly O&M costs account for 1-2% of the initial investment. Considering potential increases over the project's lifespan due to equipment wear and tear, we've opted for the higher end of this range, which is 2%. We can measure it accurately after monetizing the impacts.

9 https://www.nrel.gov/docs/fy23osti/84774.pdf
10 https://www.nrel.gov/docs/fy23osti/84774.pdf

BENEFITS

Revenue from electricity generation

The specified capacity of the wind farm, 400 MWh, is electricity to be generated in one hour. Hence, we should multiply it by the total hours in one year which is 8760 hours.

400 000 KWh x 8760 hr = 3,504,000,000 KW

However, there is no guarantee that in reality it generates electricity up to this capacity since the wind intensity is unpredictable. According to Eltel Networks, an experienced provider of infrastructure and network services, in particular electricity and telecommunications networks around the world, a 15MWh wind turbine, recently installed in Denmark, can generate 80 GW of electricity annually.[11] That is 60% of its capacity. Hence, we can assume that the expected annual electricity generation in Hecate OWF is around 2,102,400,000 KW. The wind farm is projected to produce an annual output of 2,102,400,000 KW. However, there is a loss of electricity in the transmission line, which amounts to 6.28%, according to data from BC Hydro Power.[12]. Hence, total consumable electricity is around 1,970,369,280 KW.

Social Cost of Carbon

This impact is not earned as a benefit directly from the project, but rather the shutting down of other thermal power plants which emit carbon dioxide.

According to data from US Energy Information Administration, "thermal power plants burning natural gas for electricity generation emits 0.97 pound/KWh"[13] which is 0.44 kg/KWh. Correspondingly, an annual clean

[11] https://www.eltelnetworks.pl/pl-en/blog/2023/how-much-clean-energy-does-a-wind-farm-produce/#:~:text=Energy%20production%20from%20a%20wind%20farm&text=A%20single%2C%20modern%20offshore%20wind,kWh)%20of%20electricity%20per%20year.
[12] https://www.bchydro.com/energy-in-bc/operations/transmission/transmission-scheduling/tariff-pricing/losses.html
[13] https://www.eia.gov/tools/faqs/faq.php?id=74&t=11

electricity generation of 2,102,400,000 kW (assuming that the electricity generated by gas power plants is also subject to transmission losses) can prevent the emission of 925,056 tonnes of CO_2.

Scrap value of equipment and materials at the end of the project

According to the Project Development Written Report of the Wind Farm in Wills Ridge, Floyd County, Virginia, conducted by Virginia Polytechnic Institute and State University, the scrap value of wind turbines is 7.5% of their original value.[14]

Let’s plug these numbers into our table.

[14] https://www.energy.gov/sites/prod/files/2019/06/f64/VT_SitingandProjectDevelopmentReport_2019-04-21.pdf

Types of impact	Provincial perspective					
	Costs	**Unit**	**Quantity**	**Benefits**	**Unit**	**Quantity**
Initial Inputs	Wind turbines, its substructure and foundation	Nr	33			
	Transportation	Lump sum	-			
	Installation	Nr	33			
	Electrical Infrastructure	Nr	33			
	Miscellaneous cost	Lump sum	17% of other costs			
Annuities	Cost of land	km2	55	Revenue from electricity generation	KW/ year	1,970,369,280
	Operation and maintenance	Lump sum/year	2% of the original investment	Social Cost of Carbon	Tonnes / year	925,056
Terminal Values	Decommissioning cost (At the end of the project's horizon)	MWh	400	Scrap value of equipment and materials at the end of the project	Nr	33

Step 6: Monetizing impacts

COSTS

Cost of wind turbines, its substructure and foundation.

According to the open data from Rystad Energy, 12 MW wind turbines cost around 10.1 million USD, and their foundations cost 3 to 4 million USD[15]. Consequently, a wind turbine with its foundation will be around 14.1 million USD or 19.16 million CAD.

With that in mind, the total cost of the wind turbines will be roughly **632.28 million CAD.**

Transportation costs.

The transportation costs mainly depend on the location of the manufacturer. Since most wind turbine manufacturers are based in the United States, it is evident that they will be imported from the US. Due to the substantial costs associated with transporting the massive blades and towers of these turbines, it is rational to select the nearest manufacturer with a good track record. In this regard, EDF Renewables, located in California, can be considered as the turbine supplier, as other major manufacturers are situated in the Eastern part of the continent, which could significantly escalate transportation costs.

The cost of transporting a single wind turbine for a short-haul distance is between $30,000 and $40,000 USD. For long-haul transportation, the cost can exceed $100,000 per turbine[16]. In our case, it will be a long-haul transportation even if we select the closest manufacturer. However, it's important to note that the $100,000 USD estimate is applicable to 2-3 MW wind turbines, which are evidently smaller than the 12 MW turbines we're considering. Therefore, we have decided to proportionally adjust the costs. As a result, the total transportation cost will be approximately for 33 turbines: 33 x 4 x $100,000 = $13,200,000 USD or **CAD $17,952,000**

[15] https://www.offshore-energy.biz/rystad-energy-less-is-more-if-using-14-mw-offshore-wind-turbines/
[16] https://todayshomeowner.com/eco-friendly/guides/wind-turbine-cost/

It's highly unlikely that there can be some effect on the price in the input market of transportation, thus this number reflects the total social cost.

Installation Costs.

According to Rystad Energy, the installation cost of a turbine ranges from $0.5 million to $1 million, while the cost of foundation installation ranges from $1 million to $1.5 million per unit[17]. We took the highest point of costs due to the large size of wind turbines and converted them to Canadian dollars. Thus, our calculations demonstrate these values:

Turbine installation = 33 x 1 million x 1.36 = 44.88 million CAD

Foundation installation = 33 x 1.5 million x 1.36 = 67.32 million CAD

Total installation costs = 44.88 + 67.32 = **112.2 million CAD**

We are assuming that there is a zero chance that the price of installation in the market can change due to the project.

Cost of electrical infrastructure (Grid Connection)

As outlined the step 5, the electrical infrastructure hovers around 53% of wind turbines' cost[18].

Thus, 13,700,000 x 33 x 53% = **239,613,000 CAD**

It's unlikely that the project can influence the market of spare parts for the electric infrastructure.

[17] https://w3.windfair.net/wind-energy/pr/35555-rystad-energy-size-turbine-wind-turbine-wind-farm-commercially-offshore-manufacturer-savings-costs-investments-cabling-segment
[18] https://www.nrel.gov/docs/fy23osti/84774.pdf

Cost of land.

As explained in Step 4, the opportunity cost of offshore land for human activity is almost zero. However, it might come with some externalities like damage to aquatic life, nuisances for cargo shipping, and depriving fishers of some space for hunting. However, all of these factors are pretty challenging to value. Therefore, we decided to draw a comparison with other projects.

It must be admitted that there is no single offshore wind farm in Canada to derive the cost of land for our project. We decided to estimate the cost of leasing offshore in comparison to an offshore wind farm project implemented in Rhode Island and Massachusetts by Deepwater Wind, which covers 666.7 km2 offshore land and the company paid 3,800,000 USD (5,164,390 CAD as per xe.com) and 1 km2 cost 7,746 CAD. Taking into account the considerable integrated economy of the US and Canada, the prices shouldn't vary substantially. Hence, the estimated cost of offshore land lease for the Hecate project will be as follows:

55 km2 x $7,746 CAD = **$426,030 CAD / per year**

Decommissioning Cost

According to data from the article "Modelling the decommissioning cost of offshore wind development on the U.S. Outer Continental Shelf" by Mark J. Kaiser and, Brian Snyder, the decommissioning costs are found to range from 115,000 to 135,000 $/MW in the U.S. offshore wind projects.[19]

Since the article dates back to 2012, this cost should be around 140,000 USD/MW or 189,955.43 CAD/MW

Hence, the cost of decommissioning the Hecate project is calculated as follows:

400MW x 189,955.43 CAD/MW = **75,982,172 CAD**

[19] https://doi.org/10.1016/j.marpol.2011.04.008

Miscellaneous Costs

17% of all costs of wind turbines and substructure, transportation, installation, electric infrastructure, operation and maintenance and decommission costs is calculated in the following way.

(632,280,000 + 17,952,000 + 112,200,000 + 239,613,000 + 75,982,172 + 100,564,800) x 17% = 1,178,591,972 x 17% = 200,360,635

Total miscellaneous costs are: **200,360,635 CAD**

Cost of operation and maintenance

Let's first calculate the original investment value which includes: Cost of wind turbines, its substructure and foundation, transportation costs, installation costs, cost of electrical infrastructure (Grid Connection) and miscellaneous costs which accounts for 1,102,609,800 CAD. Two percent of it is 22,052,196 CAD which is an annual O&M costs

BENEFITS

Revenue from electricity generation

In British Columbia, electricity use is billed at 13.266¢ per kilowatt hour (kWh)[20] and the total consumable electricity to be generated is around 1,970,369,280 KW. Hence, the annual revenue is projected to be **261,389,189 CAD** per year**.**

This value obviously represents a benefit to the supplier. But, can it genuinely be regarded as a social benefit if the project leads to the decommissioning of thermal power plants, consequently maintaining the total electricity supply unchanged?

[20] https://www.fortisbc.com/about-us/corporate-information/regulatory-affairs/our-electricity-utility/electric-bcuc-submissions/electricity-rates

If the amount of electricity supply in the market will not change, we can ignore this impact. We decided to leave this disputable matter to our sensitivity analysis.

Social Cost of Carbon

The latest report from the Canadian government indicates that each tonne of carbon causes $250 in damage to society.[21] As highlighted in the previous step, the project will prevent the emission of 925,056 tonnes of CO2, which can be monetized as **231,264,000 CAD a year.**

Scrap value of equipment and materials at the end of the project

As outlined above, the cost of a 12 MW wind turbine is USD 10.1 million or CAD 13.72 million. The total cost of wind turbines alone, without substructure and foundation, is CAD 452.76 million, and 7.5% of it is **CAD 33.957 million.**

Let’s put these numbers to our tables

[21] https://www.nationalobserver.com/2023/06/06/analysis/social-cost-carbon#:~:text=The%20latest%20report%20from%20the,%24250%20in%20damage%20to%20society.

Types of impacts	Provincial perspective							
	Costs	**Unit**	**Quantity**	**Monetary value**	**Benefits**	**Unit**	**Quantity**	**Monetary value**
Initial Input	Wind turbines, its substructure and foundation	Nr	33	632,280,000				
	Transportation	Lump sum	-	17,952,000				
	Installation	Nr	33	112,200,000				
	Electrical Infrastructure	Nr	33	239,613,000				
	Miscellaneous cost	Lump sum	17% of other costs	100,564,800				
Annuities	Cost of land	km2	55	426,030	Revenue from electricity generation	KW/year	1,970,369,280	261,389,189
	Operation and maintenance	Lump sum/year	2% of the original investmen	22,052,196	Social Cost of Carbon	Tonnes / ye	925,056	231,264,000
Terminal Values	Decommissioning cost (At the end of the project’s horizon)	MWh	400	75,982,172	Scrap value of equipment and materials at the end of the project	Nr	33	33,957,000

Step 7. Discounting

Since we broke down the impacts into Initial Capital, Annuities, and Terminal Values, we can calculate the impacts in present values for the whole horizon of the project by applying discounting. As we mentioned in the introduction of our CBA, the horizon of the project is 25 years and the discounting rate is 3.5%.

First, let's calculate the present value of annuities throughout the duration of the project.

		Annual Value	Total Present Value Over the Horizon of the Project
Annuities	Lease of land (One year)	426,030	7,021,620
	Operation and maintenance (One year)	22,052,196	363,453,590
	Revenue from electricity generation (One year)	261,389,189	4,308,089,733
	Social Cost of Carbon (One year)	231,264,000	3,811,580,991

Now let’s calculate the present value of Terminal Values:

Terminal Values	As per current prices	Present value at the end of the horizon of the project
Decommissioning cost (At the end of the project’s horizon)	75,982,172	32,151,627
Scrap value of equipment and materials (At the end of the project’s horizon)	33,957,000	14,368,802

The method of calculations can be found on the following link

https://docs.google.com/spreadsheets/d/1lsl3YN78Y4DAyj6FqBGNkMj_aJp4MNhQUZX2pTMCfok/edit#gid=0

Transferring these numbers to our categorized table we have the following picture:

Provincial perspective			
Costs	**Present Value**	**Benefits**	**Present Value**
Wind turbines, its substructure and foundation	632,280,000		
Transportation	17,952,000		
Installation	112,200,000		
Electrical Infrastructure	239,613,000		
Miscellaneous cost	100,564,800		
Cost of land	7,021,620	Revenue from electricity generation (conditional)	4,308,089,733
Operation and maintenance	363,453,590	Social Cost of Carbon	3,811,580,991
Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802

Step 8. Compute the net present value of each alternative

Let's first calculate the Net Present Value of the Hecate wind farm.

But we will not add the revenue generated from electricity as a benefit for now but we will provide a detailed explanation in our sensitivity analysis.

According to the table above, the total present costs of the wind farm is 1,505,236,637 and total present benefits is 3,825,949,793 (excluding the revenue generated from the electricity)

Thus, **NPV = 3,825,949,793 - 1,505,236,637 = 2,287,817,437 CAD**

To calculate the NPV of the alternative project which will be erected onshore, we should outline the main changes in the impacts. In our case, it will be an identical project in terms of capacity which is 400 MWh. The cost of construction of offshore wind farm is 20% less than onshore wind farms according to data from "Today's HomeOwner"[22] company. As specified in our calculations, construction costs comprise wind turbines, its substructure, foundation, transportation, installation, electrical infrastructure and miscellaneous costs. In total they accounted for 1,102,609,800 CAD, and applying our specified proportion for the onshore wind farm we get 918,841,500 CAD. Moreover, the operation and maintenance costs of the onshore wind farm are half of the offshore wind farms due to accessibility according to a data from "True Expertise Delivered"-Coalesce Management Consulting[23]. Hence, the O&M for the onshore wind farm is 181,726,795 CAD.

However, the opportunity cost of land onshore is much higher than offshore. According to data on Land Lease in BC from WOWA.ca, the minimum annual rent for crown land in BC is $500 per acre as of July 11, 2023[24]. We assume that the onshore land will also take up the same area of land as offshore, 55 km2 which is 13590.8 acre. Hence, the annual lease of land will cost 6,795,500 CAD. The present value of a lease of land for 25 years with 3.5 discounting rate is 112,000,132 CAD.

Another drawback of onshore wind farms is the speed and intensity of wind which is usually lower than offshore wind. In our selected area, Hecate, the wind is usually very strong and constant. For comparison, The Dokie Ridge onshore wind farm located near Chetwynd, British Columbia, has a capacity of 144 MWh, however it generates 320,000 to 340,000 gigawatt-hours of energy per year which is almost four times lower than its capacity (144 x 8760 = 1,261,440). Hence, we can infer that the onshore wind farm with 400 MWh

[22] https://todayshomeowner.com/solar/guides/onshore-vs-offshore-windpower/#:~:text=Offshore%20Wind%20Review-,Cost,of%20machinery%20out%20at%20sea.
[23] https://www.expertisedelivered.com/insights/blog/the-pros-and-cons-of-onshore-vs-offshore-wind-farms-29/
[24] https://wowa.ca/land-lease-bc

capacity will have a 25,511,111 MW output, which means the net benefit will be 1,821,448,385 CAD. Again, we will not include the benefits for now, but we will consider other benefits associated with this clean energy. Less output will have other implications on the value of other impacts, social benefit of reduced carbon emission and life saved. Their value will also change proportionally and will be as follows:

Social cost of carbon – 1,611,525,871 CAD

The decommissioning costs and scrap value of the wind farm will not vary significantly; therefore, we can keep the numbers from the offshore wind farm.

Consequently, the value of impacts for the onshore wind farm will be as follows:

Provincial perspective			
Costs	**Present Value**	**Benefits**	**Present Value**
Construction costs	918,841,500		
Cost of land	112,000,132	Revenue from electricity generation (conditional)	1,821,448,385
Operation and maintenance	181,726,795	Social Cost of Carbon	1,611,525,871
Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802
Total	**1,244,720,054**		**1,625,894,673**

As can be seen from the table above, the total present costs are 1,244,720,054 CAD and total present benefits are 1,625,894,673 CAD.

Hence, the NPV of onshore wind farm will be as follows:

NPV = 1,625,894,673 CAD - 1,244,720,054 CAD = 381,174,619 CAD

Based on the figures provided, it's evident that while the total present costs of the onshore wind farm are lower than those of the offshore wind farm, the difference in the total present value of benefits significantly favors offshore wind farms, resulting in a notably lower net present value for onshore wind farms.

It must be noted that this will happen in the scenario that the gas power plants will be decommissioned.

Considering the primary criterion for selecting a preferable project among alternatives is the net present value, it's evident that the offshore wind farm holds a considerable advantage over the onshore wind farm. Additionally, both these projects are more favourable than the status quo.

3,567,760,520 CAD > 381,174,619 CAD > 0

Step 9. Sensitivity Analysis

For our sensitivity analysis, we decided to conduct the best- and worst-case scenario sensitivity analysis since we have a few variables which might be overestimated or underestimated.

The first question raised in this Cost-Benefit Analysis (CBA) pertains to the uncertainty surrounding the construction costs. As it was stated in the 4th step of our CBA, the construction costs depend on the offer of contractors who will carry out the construction costs. The developer of the project hasn't disclosed their estimated construction and installation costs. Therefore, our calculations are solely based on the market price of wind turbines and their subcomponents. We tried to derive the cost of construction works from the other offshore wind farms which might come with some deviations, since we relied primarily on the projects implemented in the USA. Main reason for this approach is because there is no single offshore wind farm constructed in Canada. This fact itself can be a subject to discussion, how can we be sure that the construction costs can be almost identical as in the USA?

The reason why we primarily relied on the existing projects in the USA is:

First - Geographical proximity which gives a sense that most of the materials and equipment which are devoid in Canada will probably be imported from the USA. Any other option will be imported from overseas countries in either Europe or Asia. Considering that Wind turbines are large structures, it is not going to be a feasible option.

Second - The economy of these two countries are closely integrated and this levelizes the market prices in these two countries.

Nevertheless, there are some factors which might create some disparities like average wage rate and transportation.

According to www.springfinancial.ca the average salary of Americans is $56,690 and for Canada, it is about $43,867[25]. This is not in fact a significant gap and it might potentially decrease the construction costs and operation and maintenance in Canada. According to the industry-standard Construction Labor Market Analyzer (CLMA), labor cost percentages in construction lie between 20% - 40% of the total project's budget[26]. Correspondingly, the construction costs can be lowered by 4.5% - 9 %.

Regarding the transportation costs, it is logical to assume that the transportation costs are higher in Canada, since the main components will be imported from the USA. Nevertheless, the transportation costs are pretty negligible in comparison to other costs. Furthermore, we selected the highest possible transportation costs in our calculation. In this respect, the transportation costs cannot create a noticeable deviation in overall construction costs.

[25] https://www.springfinancial.ca/blog/lifestyle/is-canada-or-usa-more-expensive

[26] https://gobridgit.com/blog/how-to-calculate-construction-labor-cost/#:~:text=Construction%20labor%20costs%3A%20Fast%20facts,payroll%20taxes

Let’s analyse the construction costs of some offshore wind farms in the USA.

Skipjack Wind 2 Offshore Wind Farm with total capacity of 760 MWh is expected to cost almost $3.7 billion[27]. This wind farm is almost two times larger than the Hecate wind farm and it approximately costs almost four times more. We need to draw your attention to the fact that we excluded from the Hecate project any taxes which are usually levied from the construction works, since from CBA’s perspective it is just a transfer between the taxpayer and the government. Moreover, we don’t have precise data regarding the type of wind turbines which will be selected. Added to this, there is a chance that we are missing some inputs in our project. On top of all, as specified earlier, the construction costs are subject to a bidding process which usually includes the contractor's profit. However, considering that the profit (which can be deemed as a producer’s surplus as well) is transferred between the developer and the contractor, it cancels out anyway.

In the light of these analyses, we can deem that our construction costs are a bit optimistic.

Ideally, the best way to reflect the accuracy of our calculation is to compare the Levelized cost of electricity.

The Levelized Cost of Energy (LCOE) is the primary metric for describing and comparing the underlying economics of power projects. For wind power, LCOE represents the sum of all costs of a fully operational wind power system over the lifetime of the project, with financial flows discounted to a common year. The principal components of the LCOE for wind power systems include capital costs, operation and maintenance costs, and the expected annual energy production.

The formula to calculate the LCOE is as follows:

It - Investment expenditure in year t

Mt - Operation, maintenance and service expenditure in year t

Ft - Fuel expenditures (if applicable) (F)

[27] https://www.power-technology.com/marketdata/skipjack-wind-2-offshore-wind-farm-us-2/?cf-view

Et Energy generation in years t

r - Discount rate (or WACC), and

t - Lifetime of the project in years.

Let's calculate LCOE of Hecate project:

The investment expenditure (It), in our case includes the cost of wind turbines, transportation, installation, electrical infrastructure, lease of land for 25 years, decommissioning (without social discounting) and miscellaneous costs and the provincial sales tax.

Provincial Sales Tax (PST) applies to taxable goods used to fulfil a contract in B.C. Contractors include builders, carpenters, electricians, plumbers, painters, landscapers and anyone else who installs goods that become part of buildings or land.[28] However, we haven't found specific documents for application of PST on offshore structures in BC and whether there are special conditions.

Nevertheless, we decided to include it to our civil works cost. Generally, the rate of PST is 7% on the purchase or lease price of goods and services, with some exceptions.[29]

Our total costs of equipment, structures and infrastructure with installation costs are estimated to be 961,653,000 CAD. Consequently, estimated PST is 67,315,710 CAD

Thus, **It = 1,245,853,682**

Mt = 100,564,800

O&M growth rate - 2%

Ft - Annual fuel cost - 0

Et - Annual electricity output - 400MW x 8760 hr = 3,504,000 MW

The discount rate or WACC for Offshore Wind - 8%

T - Project's lifespan - 25 years

[28] https://www2.gov.bc.ca/gov/content/taxes/sales-taxes/pst/charge-collect/contractors

[29] https://www2.gov.bc.ca/gov/content/taxes/sales-taxes/pst#:~:text=Generally%2C%20the%20rate%20of%20PST,and%20services%2C%20with%20some%20exceptions.

Our calculation shows that LCOE of Hecate project is 0.08 CAD/KWh

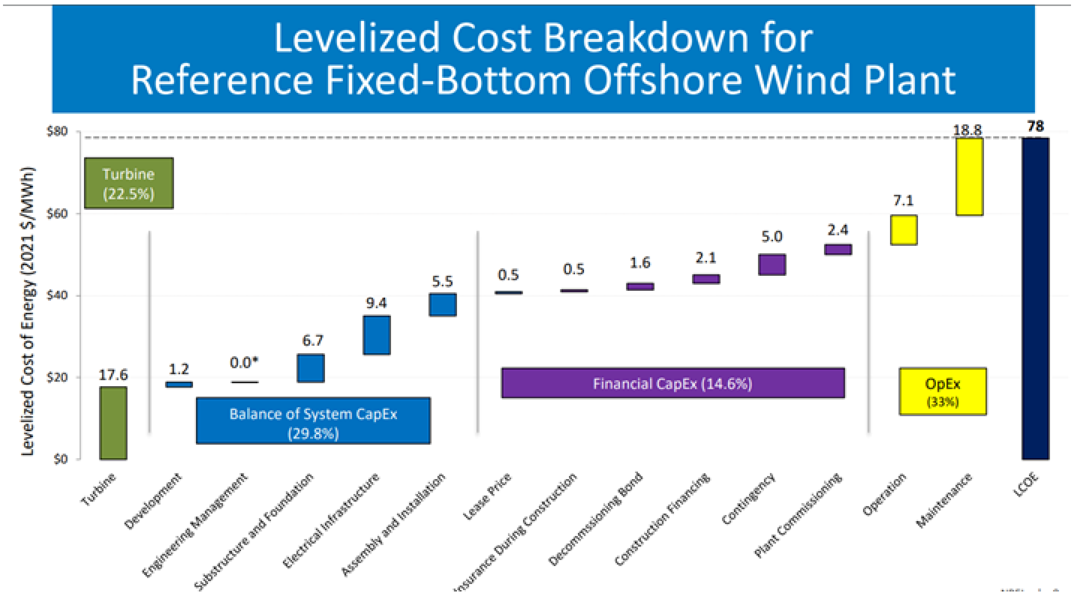
Or 80 CAD/MWh

To check our calculation, you can go to the form:

https://docs.google.com/spreadsheets/d/14rArnSBWwNtqzSr2wDCSj8o0t_VcY6HwTwKtdrm5lf8/edit#gid=1810082667

How accurate is this calculation?

Let’s compare it with the review of cost of wind energy by Tyler Stehly and Patrick Duffy in 2021



78 USD/MWh or 106.32 CAD/MWh

In this regard, we might have underestimated our construction and O&M costs by around 50%. Furthermore, the LCOE of offshore wind farms changes according to location, size, and the capacity of the plant. In the USA, the LCOE of offshore wind plants varies between $61/MWh and $116/MWh.

With that in mind, in the worst-case scenario our total construction and O&M costs will be as follows:

Construction (including wind turbines, substructures, installation, electric infrastructure, transportation, miscellaneous costs) = 1,102,555,800 x 1.5 = 1,653,833,700

O&M = 363,453,590 x 1.5 = 545,180,385

Another compelling question is whether including the revenue from the electricity output as a social benefit is a rational decision, since the project is supposed to substitute thermal power stations which are causing pollution? Considering that the total supply of electricity will remain unchanged in this scenario, it should be excluded from the benefits.

Conversely, if the project will not lead to shutting down the thermal power stations, we can include the total revenue from the electricity generation as a benefit. However, in this case, the emission of carbon dioxide will not be reduced and subsequently we will have to remove the social cost of carbon.

Considering that the social cost of carbon has lower value than the revenue from the electricity generated, the best-case scenario will be to continue using the thermal power station which also stipulates the removal of revenue from electricity generation. In the worst-case scenario, these thermal power plants will shut down and stop polluting the air which stipulates the removal of the revenue from the electricity generation from the benefits.

Let’s compute our table for the best-case scenario and the worst-case scenario for the Offshore Wind Farm Project.

The best-case scenario				The worst-case scenario			
Costs	**Present Value**	**Benefits**	**Present Value**	**Costs**	**Monetary value**	**Benefits**	**Monetary value**
Construction costs	1,102,555,800			Construction costs	1,653,555,800		
Cost of land	7,021,620	Revenue from electricity generation	4,308,089,733	Cost of land	7,021,620	Revenue from electricity generation	
Operation and maintenance	363,453,590	Social Cost of Carbon		Operation and maintenance	545,180,385	Social Cost of Carbon	3,811,580,991
Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802	Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802
Total	**1,505,182,637**		**4,322,458,535**		**2,237,909,432**		**3,825,949,793**

Hence the NPV for the best-case scenario is:

NPV = 4,322,458,535 -1,505,182,637 = 2,817,275,898 CAD

For the worst-case scenario, the NPV will be as follows:

NPV = 3,825,949,793 - 2,237,909,432 = 1,588,040,361 CAD

The method can be conducted for the onshore wind farm and the numbers look as follows:

The best-case scenario				The worst-case scenario			
Costs	**Present Value**	**Benefits**	**Present Value**	**Costs**	**Monetary value**	**Benefits**	**Monetary value**
Construction costs	918,841,500			Construction costs	1,378,262,250		
Cost of land	112,000,132	Revenue from electricity generation (conditional)	1,821,448,385	Cost of land	112,000,132	Revenue from electricity generation	
Operation and maintenance	181,726,795	Social Cost of Carbon		Operation and maintenance	272,590,193	Social Cost of Carbon	1,611,525,871
Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802	Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802
Total	**1,244,720,054**		**1,835,817,187**		**1,795,004,202**		**1,625,894,673**

According to the numbers in the table:

The NPV for the best-case scenario is:

NPV = 1,835,817,187 - 1,244,720,054 = 591,097,133 CAD

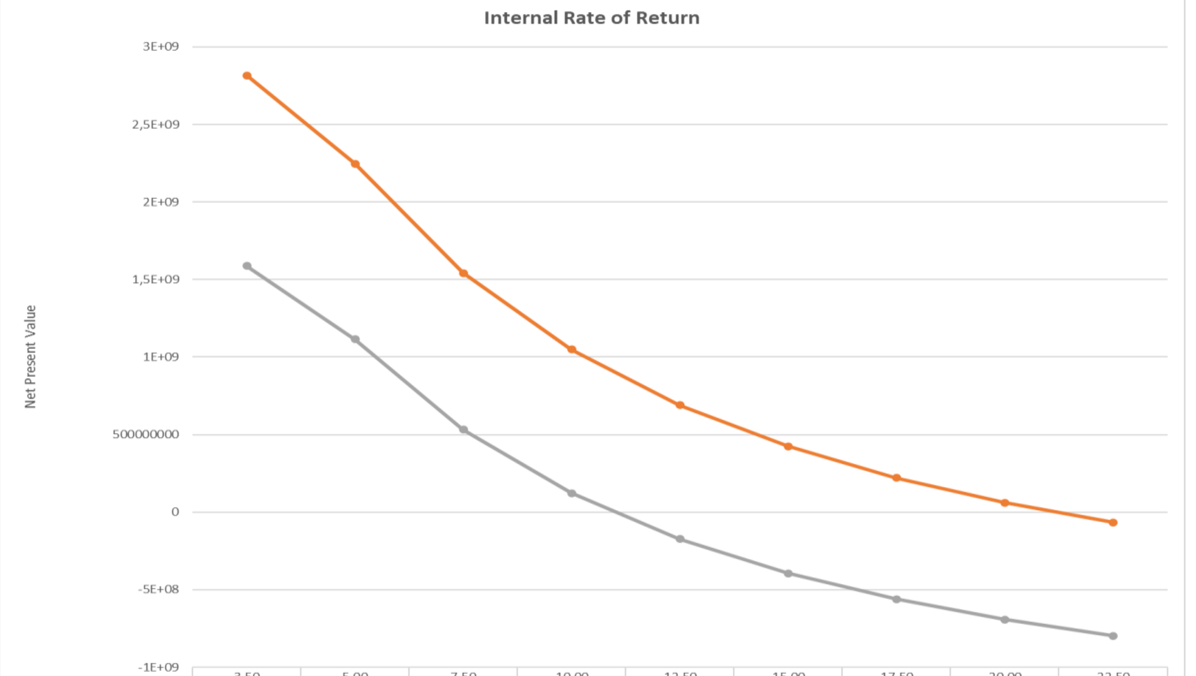
For the worst-case scenario, the NPV will be as follows:

NPV = 1625894673 - 1,795,004,202 = -169,109,529 CAD

According to these calculations we can make an inference that even in the worst-case scenario the Offshore Wind Farm project will have very high NPV, while Onshore Wind Farm seems less robust since there is possibility of having a negative NPV in the worst-case scenario. Hence, the Offshore Wind Farm project can be considered as risk free and cost-effective.

Internal Rate of Return (IRR)

Another compelling question is pertinent to the discount rate of 3.5%. As outlined in the step 7, 3.5% social discounting rate is used when the horizon of the project is less than 50 years and it is mainly used in developed countries. Even though Canada is a developed country, it's still important to consider the Internal Rate of Return. This helps determine the discounting rate at which the NPV shifts its sign.



	3,50	5,00	7,50	10,00	12,50	15,00	17,50	20,00	22,50
Best Case Scenario	2817275898	2246859244	1542788945	1048265747	690368761	424017458	220642571	61694154	-65159516
Worst Case Scenario	1588040361	1115875286	533077777	123733868	-172517049	-392990596	-561335118	-692905419	-797909142

According to this line graph we can observe the IRR of both scenarios. Under the best-case scenario, the IRR of the project is 21% and under the worst-case scenario the IRR is around 16%. Both of these rates are quite high and it is unlikely that the discounting rate can be that high in Canada which makes the project even more robust.

The calculation of IRR can be found in the following link:

https://docs.google.com/spreadsheets/d/1lsl3YN78Y4DAyj6FqBGNkMj_aJp4MNhQUZX2pTMCfok/edit#gid=0

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary and Conclusion

For an offshore wind project, a cost-benefit analysis (CBA) can offer important insights into the project's overall impact and economic sustainability. Performing a cost-benefit analysis for the Hecate Offshore Wind Farm gave us the opportunity to observe the following areas:

Financial Viability Assessment: Here, we determined the total costs associated with the offshore wind project, including initial investments, operational costs, and maintenance expenses. We evaluated the expected financial returns over the project's lifecycle, taking into account factors such as energy production, revenue from power sales, and potential government incentives.

Risk Management: We identify and assess potential risks associated with the offshore wind project, such as technical challenges, regulatory uncertainties, and market fluctuations. Develop strategies to mitigate and manage these risks, enhancing the overall resilience of the project.

Resource Allocation: Having optimised the allocation of resources, both financial and non-financial, to ensure efficiency in project execution, the investments in areas that yielded the highest returns and contributed most significantly to project success is prioritized.

Environmental and Social Impact Assessment: Our evaluation showed positive environmental impacts in generating clean energy through offshore wind, such as reduced greenhouse gas emissions and decreased reliance on fossil fuels. It also assesses potential social benefits, including job creation, community development, and improved energy security.

Regulatory Compliance: We ensured compliance with local and international regulations and standards governing offshore wind projects, then identified any regulatory hurdles and assessed their potential impact on project costs and timelines.

Stakeholder Communication: In providing a comprehensive overview of the project's costs and benefits to stakeholders, including investors, government agencies, local communities, and environmental groups. We enhanced transparency and facilitated informed decision-making among all relevant parties.

Long-Term Planning: Our analysis supports long-term planning by forecasting the economic performance of the offshore wind project over its expected lifespan (25 years) to enable stakeholders make strategic decisions regarding project expansion, upgrades, or decommissioning based on the projected costs and benefits.

Comparative Analysis: Our project compares the offshore wind project with alternative energy generation options or investment opportunities to determine its relative competitiveness and usefulness.

Financing and Investment Decisions: The analysis carried out will assist in securing project financing as it provides a clear understanding of the project's financial viability and potential returns. It will therefore facilitate investment decisions having demonstrated the project's alignment with financial goals and risk tolerance.

Policy Development: Our project will contribute valuable insights for policymakers to develop or refine energy policies that promote sustainable and economically viable offshore wind projects.

5.2 Recommendation

Step 10. Recommendations

After thoroughly scrutinising our cost-benefit analysis for the Hecate wind farm project, we can say that the project is viable and feasible. Although we faced some challenges in ascertaining the construction costs, by conducting research on other projects and some resources related to wind energy, we tried to compute the costs as feasibly as possible. There are several advantages to this offshore wind farm project, including social, environmental, and economic ones. It is evident that the primary benefit of the project is its contribution to alleviation of environmental degradation and climate change by reducing carbon dioxide without cutting overall consumption of electricity. An environmentally friendly and sustainable energy source like this, reduces greenhouse gas emissions, which helps to prevent climate change and lessen reliance on non-renewable energy sources. Even though the project will not cause the decommissioning of thermal power plants and will reduce the overall carbon emission, the electricity generated can provide even higher net present benefit than carbon reduction benefit. As outlined in the previous steps the project can reduce the pollution caused by the thermal power plants by 78.77 %.

Regarding the alternative project for building the wind farm onshore, it also demonstrated a positive NPV. However, due to the lower output of electricity on land-based wind farms in comparison to offshore wind farms, overall net present benefits are significantly lower, although construction and maintenance costs are higher in offshore wind farms.

It's worth noting even in the worst-case scenario, the NPV of the offshore wind farm will be a substantial positive number while onshore wind farm is projected to demonstrate a negative NPV.

Undoubtedly, preference should be given to the offshore wind farm because of its substantially higher NPV and less interference with nature in contrast to hydropower and thermal power. Compared to onshore settings, offshore wind farms frequently receive greater and more consistent wind speeds, increasing their potential for energy production. The increased capacity factors of offshore wind farms make the power supply more dependable and steadier. A project like this would even create more jobs in various stages, including development, construction, operation, and maintenance. This also stimulates economic growth in the surrounding regions through investments in infrastructure, services, and local businesses.

This project will be the first of its kind in not only British Columbia but also in Canada as a whole and it has a great capacity to enhance the energy security of the province and contribute to alleviation of global climate change for the whole planet. The Hecate Offshore Wind Farm can become a paragon of Canada’s energy industry and open the gate for the further development and spread of this unique technology.

References

BC Hydro (n.d.). *Losses.* https://www.bchydro.com/energy-in-bc/operations/transmission/transmission-scheduling/tariff-pricing/losses.html

This article examines a 6.28% electricity loss during transmission. Customers can mitigate this by purchasing Loss Compensation Service, reducing energy, or managing losses independently.

Beauchamp, E. (2023). *Onshore vs Offshore Wind Power*. Retrieved from https://todayshomeowner.com/solar/guides/onshore-vs-offshore-windpower/#:~:text=Offshore%20Wind%20Review-,Cost,of%20machinery%20out%20at%20sea.

This article contrasts onshore and offshore wind power, showcasing their similarities and differences. It explains that onshore wind power is presently more cost-effective compared to offshore wind. However, it predicts a future shift toward offshore wind due to decreasing costs and land scarcity, despite its current higher expenses.

Blewett, D. (2021). *Wind Turbine Cost: How Much? Are They Worth It In 2022? WeatherGuard*. https://weatherguardwind.com/how-much-does-wind-turbine-cost-worth-it/

The article explores the current costs and worth of wind turbines in 2023, discussing their expenses, electricity production, maintenance, and industry concerns about decreasing prices. It covers aspects like turbine sizes, electricity generation, operational and maintenance costs, and technological advancements.

Bridgit. (n.d.). *How to Calculate Construction Labor Cost*.

https://gobridgit.com/blog/how-to-calculate-construction-labor-cost/#:~:text=Construction%20labor%20cost s%3A%20Fast%20facts,payroll%20taxes

The article delves into efficiently calculating construction labor costs, which typically range between 20% to 40% of a project's budget. It outlines deriving base labor rates by summing individual worker wages and explores labor burden calculations involving various fees, insurance, and related expenses.

Britannica, T. Editors of Encyclopaedia (2023, November 30). hydroelectric power. Encyclopedia Britannica. https://www.britannica.com/science/hydroelectric-power

British Columbia Provincial Inventory of greenhouse gas emissions: *Current Provincial Inventory*. https://www2.gov.bc.ca/gov/content/environment/climate-change/data/provincial-inventory#:~:text=in%202021%2C%20B.C.'s%20gross,emissions%20reduction%20and%20sectoral%20targets.

This article summarizes the annual Provincial Inventory of Greenhouse Gas Emissions in British Columbia, covering emissions from 1990 onward. It also references future emission projections in BC's Provincial Forecast of Greenhouse Gas Emissions.

Buljan, A. (2020). Rystad Energy: Less Is More if Using 14 MW Turbines. *Offshore Energy*

https://www.offshore-energy.biz/rystad-energy-less-is-more-if-using-14-mw-offshore-wind-turbines/.

The article compares the financial aspects of 14 MW offshore wind turbines to smaller ones (10 MW and 12 MW), demonstrating that despite their higher initial cost, the larger turbines result in considerable savings in project expenses, notably in installation and cabling. It outlines turbine costs ranging from USD 8 million (10 MW) to USD 12.3 million (14 MW), emphasizing the expense disparity among turbine sizes.

Burger, L., Armaza, A., Clatterbuck, K., Federman, C., Maly, A., McMahon, H., Newton, E., Watt C. (2019). Virginia Tech Siting & Project Development Written Report U.S. Department of Energy Collegiate Wind Competition Virginia Polytechnic Institute and State University. https://www.energy.gov/sites/prod/files/2019/06/f64/VT_SitingandProjectDevelopmentReport_2019-04-21.pdf

The report on the Wills Ridge wind farm project in Virginia suggests it might not be financially viable due to higher-than-normal prices and limited energy generation. To improve its feasibility, suggestions include leveraging tax credits, reducing turbine costs through research, and selling Renewable Energy Credits. It also includes the salvage value of wind turbines which is around 7.5% of their original cost.

Duval, G. (2023). How Much Do Wind Turbines Cost? *Today's Homeowner*. https://todayshomeowner.com/eco-friendly/guides/wind-turbine-cost/

The article explores the expenses associated with wind turbines, detailing initial costs influenced by materials, labor, and location. It breaks down costs for manufacturing, installation, maintenance, and transportation. Overall, it underscores the long-term benefits of wind power while highlighting its environmental advantages and the necessity of renewable energy investment.

Eltel Networks. (2023). *How much clean energy does a wind farm produce?* https://www.eltelnetworks.pl/pl-en/blog/2023/how-much-clean-energy-does-a-wind-farm-produce/#:~:text=Energy%20production%20from%20a%20wind%20farm&text=A%20single%2C%20modern%20offshore%20wind,kWh

ETC/CDS. *General Environmental Multilingual Thesaurus (GEMET 2000)* European Environmental Agency. Thermal Power Plant definition. Retrieved from https://www.eea.europa.eu/help/glossary/eea-glossary/thermal-power-plant

EWEA–The European Wind Energy Association. (2010). Operation and maintenance costs of wind generated power. *Wind Energy: The Facts*. https://www.wind-energy-the-facts.org/operation-and-maintenance-costs-of-wind-generated-power.html

The article examines wind turbine operation and maintenance costs, which can start at 20-25% of the total cost per kWh and rise to 20-35% as turbines age. These costs cover insurance, maintenance, repairs, spare parts, and administrative expenses.

Feinberg, L., Jylkka, Z., Tao, M., White, B., Xu, Y., & Kendall, B. (2014). Evaluating Offshore Wind Energy Feasibility off the California Central Coast. https://tethys.pnnl.gov/sites/default/files/publications/Appendix-D-CalWindProject.pdf

This project assesses offshore wind feasibility in California's central coast for Infinity Wind and the Community Environmental Council. It involves stakeholder analysis, regulatory review, site mapping, and a benefit-cost analysis for different scenarios, highlighting financial implications and future research areas.

FortisBC (n.d.) *Electricity rates.* https://www.fortisbc.com/about-us/corporate-information/regulatory-affairs/our-electricity-utility/electric-bcuc-submissions/electricity-rates.

This piece from FortisBC's outlines the residential and commercial service charges, highlighting the bi-monthly customer charges and electricity usage costs per kilowatt-hour (kWh). The rates are specifically structured for different customer categories, including small and large commercial services.

Government of British Columbia. (2023). *Contractors and PST.* https://www2.gov.bc.ca/gov/content/taxes/sales-taxes/pst/charge-collect/contractors

The article explains how PST applies to contractors in British Columbia, detailing when they pay PST on goods used in contracts and when they charge customers.

Government of British Columbia. (2023). *Provincial Sales Tax (PST).* https://www2.gov.bc.ca/gov/content/taxes/sales-taxes/pst#:~:text=Generally%2C%20the%20rate%20of%20PST,and%20services%2C%20with%20some%20exceptions.

This publication outlines British Columbia's Provincial Sales Tax (PST) at a rate of 7% for most goods and services, although there are certain exemptions.

United States Energy Information Administration; Independence Statistics and Analysis, 2023. *How much carbon dioxide is produced per kilowatthour of U.S. electricity generation?* https://www.eia.gov/tools/faqs/faq.php?id=74&t=11.

The article details the 2022 U.S. electricity generation and resultant CO_2 emissions from utility-scale plants, emphasizing coal, natural gas, and petroleum as major contributors. It outlines the CO_2 emissions per kWh for each source and highlights the carbon-neutral nature of biomass, hydro, solar, and wind power. It also references state-level emissions data.

Hugo, C., 2023. Canada's National Observer. *The social cost of carbon* https://www.nationalobserver.com/2023/06/06/analysis/social-cost-carbon#:~:text=The%20latest%20report%20from%20the,%24250%20in%20damage%20to%20society

The piece explores the social cost of carbon, indicating that each tonne of carbon creates $250 in societal damage. It delves into policy implications and the need to align carbon pricing with this cost, emphasizing the unsustainability of certain lifestyle choices.

Investopedia (n.d). Cost of Capital vs. Discount Rate: What's the Difference? (2022). Retrieved from https://www.investopedia.com/ask/answers/052715/what-difference-between-cost-capital-and-discount-rate.asp#:~:text=The%20discount%20rate%20is%20the%20interest%20rate%20used%20to%20calculate,budgeting%20for%20a%20new%20project.

Investopedia (n.d.). Net Present Value (NPV) definition. Retrieved from https://www.investopedia.com/terms/n/npv.asp#:~:text=Net%20present%20value%20(NPV)%20is%20used%20to%20calculate%20the%20current,minimum%20acceptable%20rate%20of%20return.

Kaiser, M. J., & Snyder, B. (2012). Modeling the decommissioning cost of offshore wind development on the U.S. Outer Continental Shelf. *Marine Policy*, *36*(1), 153-164. https://doi.org/10.1016/j.marpol.2011.04.008.

The paper estimates the decommissioning costs for U.S. offshore wind farms, focusing on factors like removal, processing, and disposal expenses. Estimated costs range from $115,000 to $135,000 per MW, comprising about 3-4% of the total capital costs for these projects.

Lauraine, G. C. Stratus Consulting Inc. & Paul D& Paullicy Research, 2009. Economic Valuation of Mortality Risk Reduction: Review and Recommendations for Policy and Regulatory Analysis Research Paper. https://publications.gc.ca/collection_2009/policyresearch/PH4-51-2009E.pdf.

The article examines how policymakers value reductions in mortality risks in Canada. It delves into willingness-to-pay (WTP) measures, like the value of statistical life (VSL), which help estimate the cost-benefit of policies. It emphasizes the complexities in these estimations and the need for more research in this field.

Moore, M. A., Boardman, A. E., Vining, A. R., Weimer, D. L., & Greenberg, D. H. (2004). Just Give Me a Number! Practical Values for the Social Discount Rate. *Journal of Policy Analysis and Management, 23*(4), 789–812.

The article provides specific guidelines for choosing the social discount rate based on project characteristics, aiming to standardize its use in cost-benefit analysis. It suggests different rates for intragenerational and intergenerational projects.

Mogul, F. (2019). *Offshore Wind May Help The Planet — But Will It Hurt Whales*? NPR. https://www.npr.org/2019/12/05/782694371/offshore-wind-may-help-the-planet-but-will-it-hurt-whales

Office of the US Energy Efficiency & Renewable Energy (n.d.). *How Do Wind Turbines Work?* https://www.energy.gov/eere/wind/how-do-wind-turbines-work

Oxford Learner's Dictionaries. (n.d.). Definition of Offshore. Retrieved from https://www.oxfordlearnersdictionaries.com/definition/english/offshore_1?q=offshore+

Oxford Learner's Dictionaries. (n.d.). Definition of Wind farm. Retrieved from https://www.oxfordlearnersdictionaries.com/definition/english/wind-farm

Oxford University Press (n.d.) Decommissioning Cost definition. Retrieved from: https://www.oxfordreference.com/display/10.1093/oi/authority.20110803095706141

Power Technology. (2021). *Skipjack Wind 2 Offshore Wind Farm*

https://www.power-technology.com/marketdata/skipjack-wind-2-offshore-wind-farm-us-2/?cf-view

The article provides details about the Skipjack Wind 2 Offshore Wind Farm, a 760MW project planned in Maryland, USA, developed by Orsted US Offshore Wind and is expected to power 250,000 households upon completion in 2027, with an estimated cost of $3.7 billion.

Spring Financial. (2023, July 14). *Is Canada or the USA more Expensive?* https://www.springfinancial.ca/blog/lifestyle/is-canada-or-usa-more-expensive.

The article compares the cost of living in Canada and the USA, discussing healthcare, taxes, job prospects, and living standards. It explains that the USA often offers higher salaries, Canada generally has lower housing and consumer costs.

Stanley, A. (2018). *Mapping the US-Canada energy relationship*. Center for Strategic & International Studies.

https://www.csis.org/analysis/mapping-us-canada-energy-relationship.

The report analyses the energy trade between the U.S. and Canada, showing shared priorities and using a map to illustrate the relationship.

Stehly, T., & Duffy. (2021). 2020 Cost of Wind Energy Review (No. NREL/TP-5000-81209) (p. 8). National Renewable Energy Lab (NREL), Golden, CO (United States). https://www.nrel.gov/docs/fy23osti/84774.pdf

This article examines the levelized cost of energy (LCOE) for both land-based and offshore wind power plants in the United States, using 2021 data. It aims to understand cost variability and includes estimated LCOE for various wind projects, and sensitivity analyses.

Thomson Environmental Consultants. (2021). *Offshore wind decommissioning*. Retrieved from: https://www.thomsonec.com/news/offshore-wind-decommissioning/#:~:text=Eventually%2C%20all%20windfarms%20will%20reach,turbines%20and%20removing%20the%20structure.

The article explores the complex decisions involved in decommissioning offshore wind farms nearing the end of their lifespan. It outlines three main options: extending the life, upgrading components (repowering), or complete decommissioning.

Timblin, D. (2022). *The Pros and Cons of Onshore vs Offshore Wind Farms*. Retrieved from https://www.expertisedelivered.com/insights/blog/the-pros-and-cons-of-onshore-vs-offshore-wind-farms-29/

This piece delves into the comparison between onshore and offshore wind farms, discussing their advantages and drawbacks. It explores cost-effectiveness, installation speed, economic impact, and environmental considerations for both types. Ultimately, it weighs the pros and cons, foreseeing offshore wind as a potential future alternative despite its current higher expenses and maintenance challenges.

United States Energy Information Administration; Independence Statistics and Analysis, 2023. How much carbon dioxide is produced per kilowatthour of U.S. electricity generation? Wind energy. (2023). *International Renewable Energy Agency (IRENA)*

https://www.irena.org/Energy-Transition/Technology/Wind-energy

The article emphasizes wind energy's expansion, technological progress, and reduced costs. It highlights the substantial growth in installed capacity, notably offshore, mentioning the increased size of modern turbines onshore and offshore.

Windfair. (2020). *Size matters in offshore wind: Why costlier 14 MW turbines actually reduce the large-scale farm bill.* https://w3.windfair.net/wind-energy/pr/35555-rystad-energy-size-turbine-wind-turbine-wind-farm-commercially-offshore-manufacturer-savings-costs-investments-cabling-segment

The article compares the financial gains of 14 MW offshore wind turbines against smaller 10 MW and 12 MW ones. It shows that despite pricier manufacturing, opting for 14 MW turbines results in substantial savings for large-scale projects. It also highlights lowered foundation and installation costs, making offshore wind farms more financially feasible overall.

Wind Measurement International. (n.d). *Operational and Maintenance Costs for Wind Turbines.* http://www.windmeasurementinternational.com/wind-turbines/om-turbines.php

WOWA. (2023, July 11). *Long-Term Residential Lease in BC*. Retrieved from https://wowa.ca/land-lease-bc

This article explores leasehold ownership in Canada, focusing on land lease houses and the types of land available for leasing in British Columbia. It details the advantages and disadvantages of leasehold homes, emphasizing cost considerations, property rights, and land lease tenure lengths.

APPENDIX

1. CALCULATION FOR LEVELIZED COST OF ENERGY

Let's compare it with the review of cost of wind energy by Tyler Stehly and Patrick Duffy in 2021

It = 1,245,853,682

Mt = 100,564,800

O&M growth rate - 2%

Ft - Annual fuel cost - 0

Et - Annual electricity output - 400MW x 8760 hr = 3,504,000 MW

The discount rate or WACC for Offshore Wind - 8%

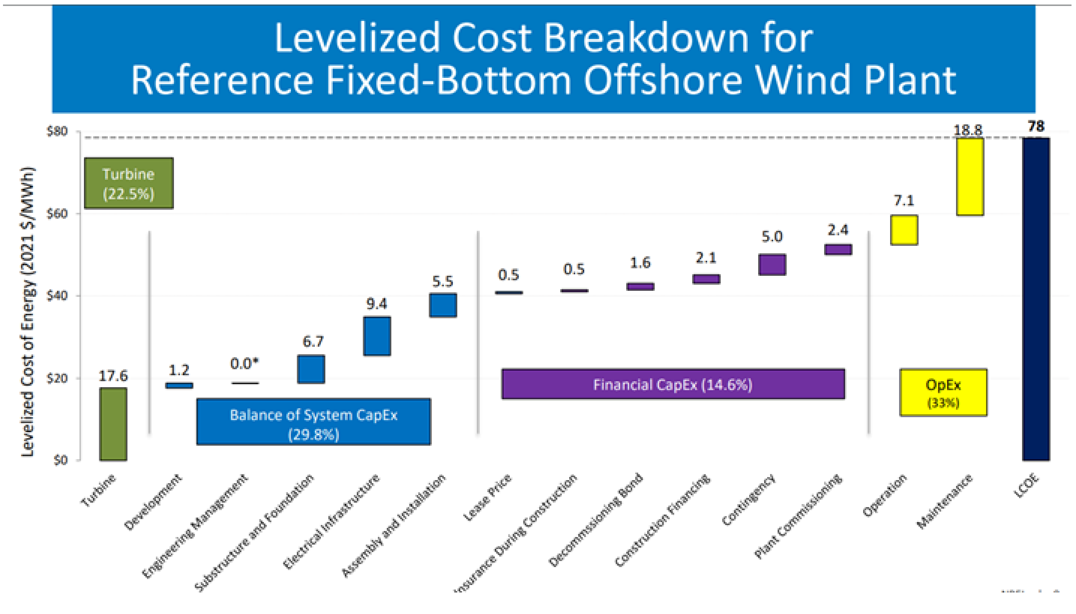
T - Project's lifespan - 25 years

Our calculation shows that LCOE of Hecate project is 0.08 CAD/KWh

Or 80 CAD/MWh

To check our calculation, you can go to the form: https://docs.google.com/spreadsheets/d/14rArnSBWwNtqzSr2wDCSj8o0t_VcY6HwTwKtdrm5lf8/edit#gid=1810082667

How accurate is this calculation?



2. DISCOUNTING CALCULATION

The calculation of IRR can be found in the following link:

https://docs.google.com/spreadsheets/d/1lsl3YN78Y4DAyj6FqBGNkMj_aJp4MNhQUZX2pTMCfok/edit#gid=0

Levelized Cost of Energy Template (LCOE)

Assumptions (in '000s)	
Initial Investment Cost ($)	1,245,853,682
Operations and Maintenance Costs ($)	22,052,196
O&M Growth Rate (%)	2.00%
Annual Fuel Costs ($)	-
Annual Electricity Output (kWH)	1,970,369,280
Project Lifespan (years)	25
Discount Rate (%)	8.00%
Entry Date	1/10/2023

Total Costs	**Entry**	Construction	Operations
Date	1/10/2023	1/10/2024	1/10/2025
Year Frac (From Start Date)		1	2
Initial Investment	1,245,853,682	-	-
O&M Costs	-	-	22,052,196
Fuel Costs	-	-	-
Discount Factor		0.926	0.857
Present Value of Costs	1,245,853,682	-	18,906,204
NPV of Total Costs	**$1,499,845,469**		

Total Energy Output	Entry	1	2
Yearly Output	-	-	1,970,369,280
Discount Factor	0.000	0.926	0.857
Present Value of Costs	-	-	1,689,274,074
NPV of Total Output	**19,208,835,073kWh**		

LCOE	**$0.08/kWh**

78.08102172

Operations	Operations	Operations	Operations	Operations	Operations
1/10/2026	1/10/2027	1/10/2028	1/10/2029	1/10/2030	1/10/2031
3	4	5	6	7	8
-	-	-	-	-	-
22,493,240	22,943,105	23,401,967	23,870,006	24,347,406	24,834,354
-	-	-	-	-	-
0.794	0.735	0.681	0.630	0.583	0.540
17,855,859	16,863,867	15,926,985	15,042,153	14,206,478	13,417,229

3	4	5	6	7	8
1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280
0.794	0.735	0.681	0.630	0.583	0.540
1,564,142,661	1,448,280,242	1,341,000,224	1,241,666,874	1,149,691,550	1,064,529,213

14.7

Operations	Operations	Operations	Operations	Operations	Operations
1/10/2032	1/10/2033	1/10/2034	1/10/2035	1/10/2036	1/10/2037
9	10	11	12	13	14
-	-	-	-	-	-
25,331,041	25,837,662	26,354,416	26,881,504	27,419,134	27,967,517
-	-	-	-	-	-
0.500	0.463	0.429	0.397	0.368	0.340
12,671,827	11,967,837	11,302,957	10,675,015	10,081,959	9,521,850

9	10	11	12	13	14
1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280
0.500	0.463	0.429	0.397	0.368	0.340
985,675,197	912,662,220	845,057,611	782,460,751	724,500,695	670,833,977

Operations	Operations	Operations	Operations	Operations	Operations
1/10/2038	1/10/2039	1/10/2040	1/10/2041	1/10/2042	1/10/2043
15	16	17	18	19	20
-	-	-	-	-	-
28,526,867	29,097,404	29,679,352	30,272,939	30,878,398	31,495,966
-	-	-	-	-	-
0.315	0.292	0.270	0.250	0.232	0.215
8,992,858	8,493,255	8,021,407	7,575,774	7,154,897	6,757,403

15	16	17	18	19	20
1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280
0.315	0.292	0.270	0.250	0.232	0.215
621,142,571	575,132,010	532,529,639	493,082,999	456,558,333	422,739,197

Operations	Operations	Operations	Operations	Operations
1/10/2044	1/10/2045	1/10/2046	1/10/2047	1/10/2048
21	22	23	24	25
-	-	-	-	-
32,125,886	32,768,403	33,423,771	34,092,247	34,774,092
-	-	-	-	-
0.199	0.184	0.170	0.158	0.146
6,381,992	6,027,437	5,692,579	5,376,325	5,077,640

21	22	23	24	25
1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280	1,970,369,280
0.199	0.184	0.170	0.158	0.146
391,425,182	362,430,724	335,584,004	310,725,930	287,709,194

Horizaon of the Project (years)	25
Discounting rate	0.035

Years of the project

Annuities	Annual Values	1	2	3	4	5	6	7	8	9	10	11	12
Lease of land offshore (One year)	426,030	411623.1884	397703.5637	384254.6509	371260.5323	358705.8283	346575.6795	334855.729	323532.1053	312591.4061	302020.6822	291807.4224	281939.5386
Operation and maintenance (One year)	22,052,196	21306469.57	20585960.93	19889817.33	19217214.81	18567357.3	17939475.65	17332826.72	16746692.48	16180379.21	15633216.63	15104557.13	14593775
Revenue from electricity generation (One year)	261,389,189	252549941.1	244009604.9	235758072.4	227785577.2	220082683.2	212640273.7	205449539.8	198501970.8	191789343.8	185303713.8	179037404.6	172982999.6
Social Cost of Carbon (One year)	231,264,000	223443478.3	215887418.6	208586877.9	201533215.3	194718082.5	188133413	181771413.5	175624554.1	169685559.6	163947400.5	158403285.5	153046652.7
Lease of land onshore (C	6,795,500	6565700.483	6343671.964	6129151.656	5921885.658	5721628.655	5528143.628	5341201.573	5160581.23	4986068.822	4817457.799	4654548.598	4497148.404

Annuities	13	14	15	16	17	18	19	20	21	22	23	24	25	Total Present Value Over the Horizon of the Project
Lease of land offshore (One year)	272405.3513	263193.5761	254293.3103	245694.0196	237385.5262	229357.9963	221601.9288	214108.1437	206867.7717	199872.2432	193113.2785	186582.8778	180273.3118	**7021619.662**
Operation and maintenance (One year)	14100265.7	13623445.12	13162748.91	12717631.8	12287566.96	11872045.37	11470575.23	11082681.39	10707904.72	10345801.66	9995943.636	9657916.556	9331320.344	**363453590.2**
Revenue from electricity generation (One year)	167133333	161481481.1	156020754.7	150744690.6	145647044	140721781.7	135963074.1	131365289	126922984.5	122630902.9	118483964.1	114477260	110606048.4	**4308089733**
Social Cost of Carbon (One year)	147871162	142870687.9	138039312	133371316	128861174.8	124503550.6	120293285.6	116225396.7	112295069.3	108497651.5	104828648.8	101283718.6	97858665.33	**3811580991**
Lease of land onshore (C	4345070.921	4198136.156	4056170.199	3919005.023	3786478.283	3658433.124	3534717.994	3415186.468	3299697.07	3188113.111	3080302.523	2976137.703	2875495.366	**112000132.4**

Terminal Impacts	As per current prices	Present value at the end of the horizon of the project
Decommissioning cost (At the end of the project's horizon)	75,982,172	**32151627.32**
Scrap value of equipment and materials (At the end of the project's horizon)	33,957,000	**14368802.31**

The best case scenario				The worst case scenario			
Costs	**Present Value**	**Benefits**	**Present Value**	**Costs**	**Monetary value**	**Benefits**	**Monetary value**
Construction costs	1,102,555,800			Wind turbines, its substructure and foundation	1,653,555,800		
Cost of land	7,021,620	Revenue from electricity generation	4,308,089,733	Cost of land	7,021,620	Revenue from electricity generation	
Operation and maintenance	363,453,590	Social Cost of Carbon		Operation and maintenance	545,180,385	Social Cost of Carbon	3,811,580,991
Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802	Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802
Total	**1,505,182,637**		**4,322,458,535**		**2,237,909,432**		**3,825,949,793**

	IRR	
discount rate	Best Case	Worse Case
3.5	2,817,275,898	1,588,040,361
5	2,246,859,244	1,115,875,286
7.5	1,542,788,945	533,077,777
10	1,048,265,747	123,733,868
12.5	690,368,761	-172,517,049
15	424,017,458	-392,990,596
17.5	220,642,571	-561,335,118
20	61,694,154	-692,905,419
22.5	-65,159,516	-797,909,142

Provincial perspective				The worst case scenario			
Costs	**Present Value**	**Benefits**	**Present Value**	**Costs**	**Monetary value**	**Benefits**	**Monetary value**
Construction costs	918,841,500			Construction costs	1,378,262,250		
Cost of land	112,000,132	Revenue from electricity generation (conditional)	1,821,448,385	Cost of land	112,000,132	Revenue from electricity generation	
Operation and maintenance	181,726,795	Social Cost of Carbon		Operation and maintenance	272,590,193	Social Cost of Carbon	1,611,525,871
Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802	Decommissioning cost (At the end of the project's horizon)	32,151,627	Scrap value of equipment and materials at the end of the project	14,368,802
	1,244,720,054		1,835,817,187		1,795,004,202		1625894673
		591,097,133				-169,109,529	